📊 Sales Forecast Analysis: LOC Count, Comparison, and ETA Estimation

Sales Forecast (SalesForecast-main) — LOC Count, Comparison, and ETA

Document type: Complexity assessment + man-hour estimation (same method as Comparison ETA – PAM)
Module: Sales Forecast / Sales Projections (`SalesForecast.exe` , `prjSalesForecast`)
Source scanned: `C:\\Users\\akshara.karikalan\\Downloads\\SalesForecast`
Benchmarks:

- Project Calendar — `C:\\Users\\akshara.karikalan\\Downloads\\ProjectCalendar` + `Calendar.txt` (**860 hours**, index **1.0**)
- Scheduling Board — `C:\\Users\\akshara.karikalan\\Downloads\\SchedulingBoard-main` + `Scheduling-board.txt` (**1,480 hours**, index **1.72**)
- Method example — `Comparison+ETA+-+PAM.doc` (PAM = **800 hours**, index **0.93**)
  Date: 16 Sep 2026
  Stack of current code: Visual Basic 6 desktop EXE (not .NET). Estimate below is for a **Finesse-style .NET WinForms modernization**, matching how Calendar / Scheduling Board / PAM were estimated.

A. Lines of code — necessary application lines only

Counting rules (what “necessary” means)

Included:

- Executable product logic (filters, SQL/SP call, tree render, Excel export, tool launch, settings)
- Helper code **actually called** by that logic
- `ColumnTreeViewStateHelper.cls` (tree expand/scroll restore)
  Excluded:
- Visual SourceSafe / IDE junk (`.scc` , `.vbw` , `.log`)
- Binary resources (`.frx`)
- VB6 form **designer layout** (control coordinates; needed to compile, not business logic)
- Blank lines and comment-only lines
- Dead handlers (`optRevenueGross_Click` / `optRevenueNet_Click` — controls are not on the form)

- **Unused** `Utilities.bas` routines (shared Finesse dump; same file also appears in Project Calendar)
- Duplicate `Utilities.AppendSQL` (the form has its own `AppendSQL`)

A.1 File-level count

File	Gross lines	Designer / layout	Blank	Comments	Logic (non-blank, non-comment)	Necessary application logic
`frmSalesForecast.frm`	2,709	470	522	128	1,589	**1,581** (minus ~8 dead option-button handlers)
`ColumnTreeViewStateHelper.cls`	101	0	17	4	80	**80**
`General.bas`	5	0	2	0	3	**3** (`Global cnApp`)
`Utilities.bas` (entire file)	903	0	189	47	667	**~185** used subset only
`prjSalesForecast.vbp`	48	—	—	—	—	Project file, not app logic

`.scc` / `.vbw` / `.log` / `.frx`	—	—	—	—	—	**0** (not application code)
Total necessary application LOC						**1,849**

Necessary `Utilities.bas` subset (used by this EXE)

Used symbol	Why it is needed
`SafeItemData`, `MatchComboText`, `MatchComboItemData`	Filter combo selection
`SetComboBoxHeight` + `SendMessageLong`, `GetWindowRect`, `ScreenToClient`, `MoveWindow`, `POINTAPI`, `RECT`, `CB_GETITEMHEIGHT`	Dropdown height
`SaveWindowState`, `LoadWindowState`	Window geometry persistence
`SysColorToRGB`, `GetRValue`, `GetGValue`, `GetBValue`, `OleTranslateColor`	Tree/Excel colors
`CreateProcess`, `STARTUPINFO`, `PROCESS_INFORMATION`, `STARTF_USESHOWWINDOW`, `NORMAL_PRIORITY_CLASS`, `SW_SHOWNORMAL`	Project Maintenance / Project Detail launch

Not counted in the 1,849 (present in `Utilities.bas` but unused here): clipboard retry, timezone conversion, `BringWindowToFront`, FlexGrid autosize, SQLQuote/SQLDate, RGBA helpers, `RunningInIDE`, etc. (~480 logic lines of shared leftover).

A.2 Headline LOC numbers (use these)

Measure	Lines	Use
Necessary application LOC	1,849	Effort / complexity (this number)
UI layout (VB6 designer in `.frm`)	470	Compile-time form definition only
Necessary logic + UI layout	2,319	If you count screen definition as "app"
All `.frm/.bas/.cls` including unused Utilities	3,718	Inflated; do not use for estimation
Whole folder including SCC/IDE/binary	n/a	Not source of product behavior

SalesForecast-main necessary application lines = 1,849.

B. Comparison vs Project Calendar and Scheduling Board

B.1 Necessary application LOC (same rules)

Module	Runtime today	App logic LOC (excl. unused shared Utilities / excl. designer)	Designer / WinForms layout	Active stored procs in client	Forms	Writes business data?
Sales Forecast	VB6 EXE	**1,849**	470	1 (`get_sales_forecast_quoted_by_warehouse`, 14 recordsets)	1	No
Project Calendar	VB6 EXE	**4,406** (excl. same `Utilities.bas` dump)	2,388	Multiple (calendar/notes/crew)	5 (`frmProjCal`, notes, crew, company picker,	Yes (notes CRUD)

					warehouse picker)	
Scheduling Board	VB Shop 4.8 WinForms	**4,967** (Forms + module only; excl. ClairUtilities library)	2,130	11	3	Yes (substitution, warehouse groups, commit)

LOC ratios vs Calendar logic (4,406):

Module	Logic LOC / Calendar	Hours benchmark	Hours / Calendar
Project Calendar	1.00	860	1.00
Sales Forecast	**0.42**	(this estimate)	**0.64**
Scheduling Board	1.13 (app forms only; real scope much higher with ClairUtilities + 3 views)	1,480	1.72
PAM (from example doc)	n/a	800	0.93

Hours do **not** scale 1:1 with LOC. Sales Forecast is smaller, but still pays a floor cost for QA permutations, Excel export, SP contract, and Finesse EXE launches. That is why the hour index (0.64) is higher than the LOC index (0.42).

B.2 Scope comparison (from code + Calendar.txt + Scheduling-board.txt)

| Capability | Sales Forecast | Project Calendar (860h) | Scheduling Board (1,480h) |
| --- | --- | --- | --- |
| Primary UX | One hierarchical tree, 12-month money grid | Weekly calendar grid + color states | Three views (Board / Timeline / Calendar), merged cells, frozen columns |
| User workflows | Filter → query → view / Excel / launch tool | Navigate weeks, notes CRUD, crew inquiry, context menus | Availability, substitution/split, warehouse groups, crew mode |
| Edits to SQL business data | None | Notes insert/update/delete | Substitution commit,

warehouse group commit, sequence updates |
Filter surface	WH, 2 AEs, company, group, dates, status, proposal type, probability, currency	Company + multi-warehouse, week, note filters	Parts, warehouses/groups, dates, proposals, summarize, crew divisions
Display modes	Super-user hierarchy vs flat	Multi-warehouse prefixes; crew dialog	3 full view modes + detail panes
Excel	Tree snapshot with grouping/colors	Crew inquiry export	Grid export
Sibling EXEs	4 (Proj Maint, Bid Report, Quote Report, Inv vs Bid)	Several (context-menu out-of-module actions)	9+
Auto-refresh	No	Yes, configurable	Manual + delayed refresh
AuthZ in client	SQL role `SalesForecastSuperUsers` + `my_company_visible`	User prefs / warehouse-company	Warehouse visibility, groups, crew
UI density	Medium-Low	Medium	High
Backend depth	Medium (1 fat SP, 14 RS mapping)	Medium (many queries + notes writes)	High (availability math, 11 SPs)
Classification vs mandated labels:
Less complex than Calendar.txt
(and far less complex than Scheduling-board.txt; also less complex than PAM)

C. Estimation (PAM document structure)

1. Functional coverage checklist

All **active** Sales Forecast behaviors from the VB6 code are included in the estimate.

Disabled/commented items are listed as out of scope unless product asks to revive them.

Active features (in estimate)

- [x] F-01 Startup, SQLOLEDB connect via `ESS*` env vars
- [x] F-02 Super-user flag (`SalesForecastSuperUsers`)
- [x] F-03 Account Executive + 2nd AE combos (`ACCTEXEC`)
- [x] F-04 Currency combo; default from `my_user_info`
- [x] F-05 Probability threshold 0/25/50/75/100
- [x] F-06 Warehouse combo
- [x] F-07 Billing company combo + `(all companies; flat)` for super-user
- [x] F-08 Company group combo
- [x] F-09 12-month window, year−2…year+2, quarter/year buttons
- [x] F-10 Project status list (exclude `I`)
- [x] F-11 Proposal types (`DoJobCosting = 1`)
- [x] F-12 Restore/save filters + window (registry)
- [x] F-13 Async `get_sales_forecast_quoted_by_warehouse`
- [x] F-14 Tree render: company / parent / project×warehouse, thousands, colors, RFI actuals, top line, grand totals
- [x] F-15 Filter auto-refresh

- [x] F-16 Linked start/end months
- [x] F-17 Expand/collapse; hide parent dollars when expanded
- [x] F-18 Show/hide AE/Ops columns
- [x] F-19 Excel export (group, colors, thousands, outline)
- [x] F-20 Launch Project Maintenance
- [x] F-21 Launch Project Detail (requires project #)
- [x] F-22 Launch Bid Report
- [x] F-23 Launch Quote Report
- [x] F-24 Launch Invoicing vs Bid
- [x] F-25 Unload persistence
- [x] F-26 Resize + End

  Active feature count: 26 (vs PAM 35, Calendar ~8 feature groups / many stories, Scheduling Board much larger)

Explicitly not in estimate (gaps / leftovers in current code)

- Combined Grand Total / Actual Grand Total (commented out)
- Alternate SPs (`get_sales_forecast` , `quoted` , `quoted_vs_manual`)
- Gross vs Net option buttons (dead)
- Rewriting `get_sales_forecast_quoted_by_warehouse` internals (SP not in repo) — **risk adder only**
- React / .NET 8 Web API rebuild (different delivery model than PAM/SB/Calendar WinForms estimates)

Business rules included

BR-01 to BR-19 from the as-is FSD (12-month window, thousands display, probability ≥, status/proposal filters, super-user layout, color rules, read-only SQL, etc.)

2. Complexity breakdown

Area	Level	Justification
UI / Frontend	**Low–Medium**	One form, many filters, one tree. No notes UI, no 3-view board, no calendar cell engine. Excel grouping/colors add work. Simpler than Calendar; much simpler than Scheduling Board.

Backend & API	**Medium**	One stored procedure but **14 recordsets** and strict `NextRecordset` order. Lookup SQL + role/visibility. No INSERT/UPDATE. Simpler than PAM writes and SB availability math. **SP body unknown** (risk).
Workflow & orchestration	**Low**	Filter → async fetch → render. No attach/detach, no substitution, no notes CRUD.
Data model & persistence	**Low–Medium**	Registry settings only on the client. Reads Company, Warehouse, Currency, ProjectStatus, pjproptype, employee job types, SysConfig. Forecast grain lives in the SP.
Integration & dependencies	**Low–Medium**	Excel COM + 4 sibling EXEs + env SQL login. Fewer than Calendar context menus and far fewer than SB's 9+ tools.